EXPERIMENT #31

Timestamp: 20260323_131752 | Environment: Local

1. Experiment Parameters & Network Statistics

Parameter	Value
Number of sticks (N)	30
Stick length (a)	0.7
Domain size	1.0 × 1.0
Source region	x < 0.1
Drain region	x > 0.9
Total intersections	196
Has percolation	✓ Yes
# Percolating paths	1
Nodes before pruning	196
Nodes after pruning	3
Edges before pruning	195
Edges after pruning	2

2. Network Visualizations

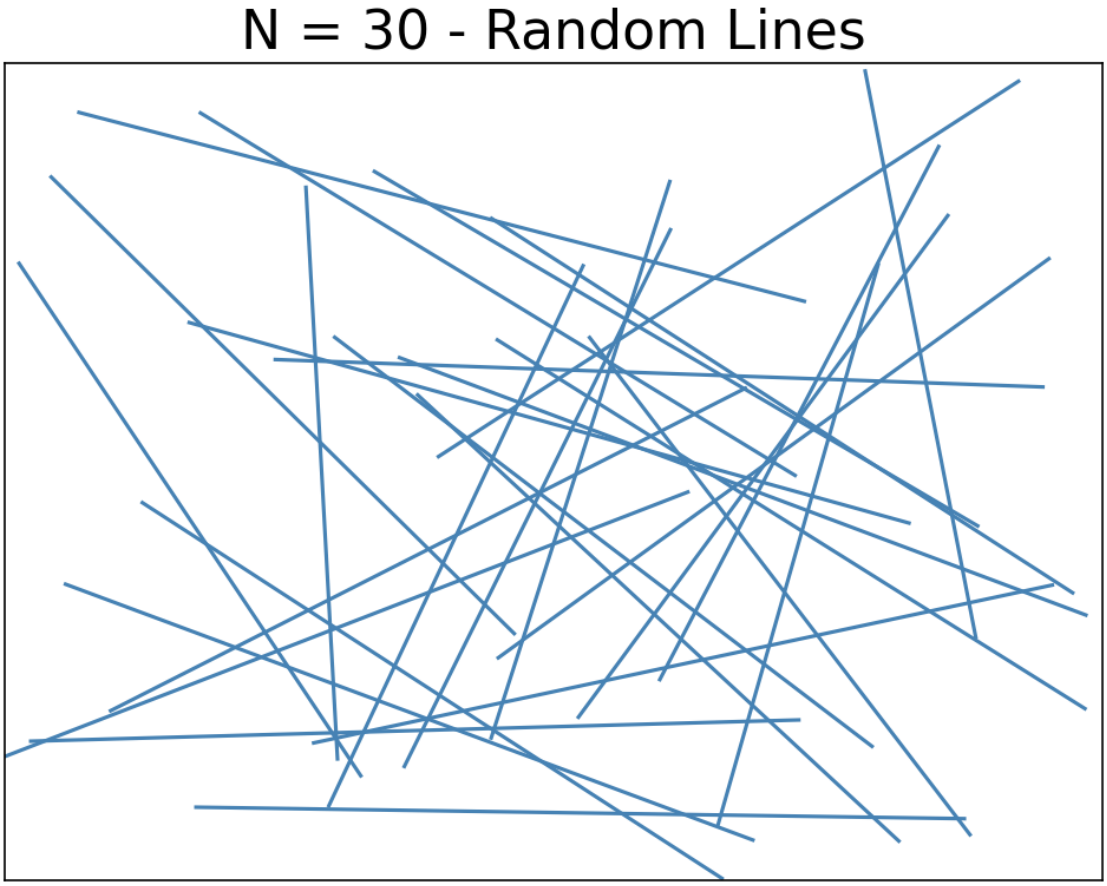


Fig. — 01 random lines N30

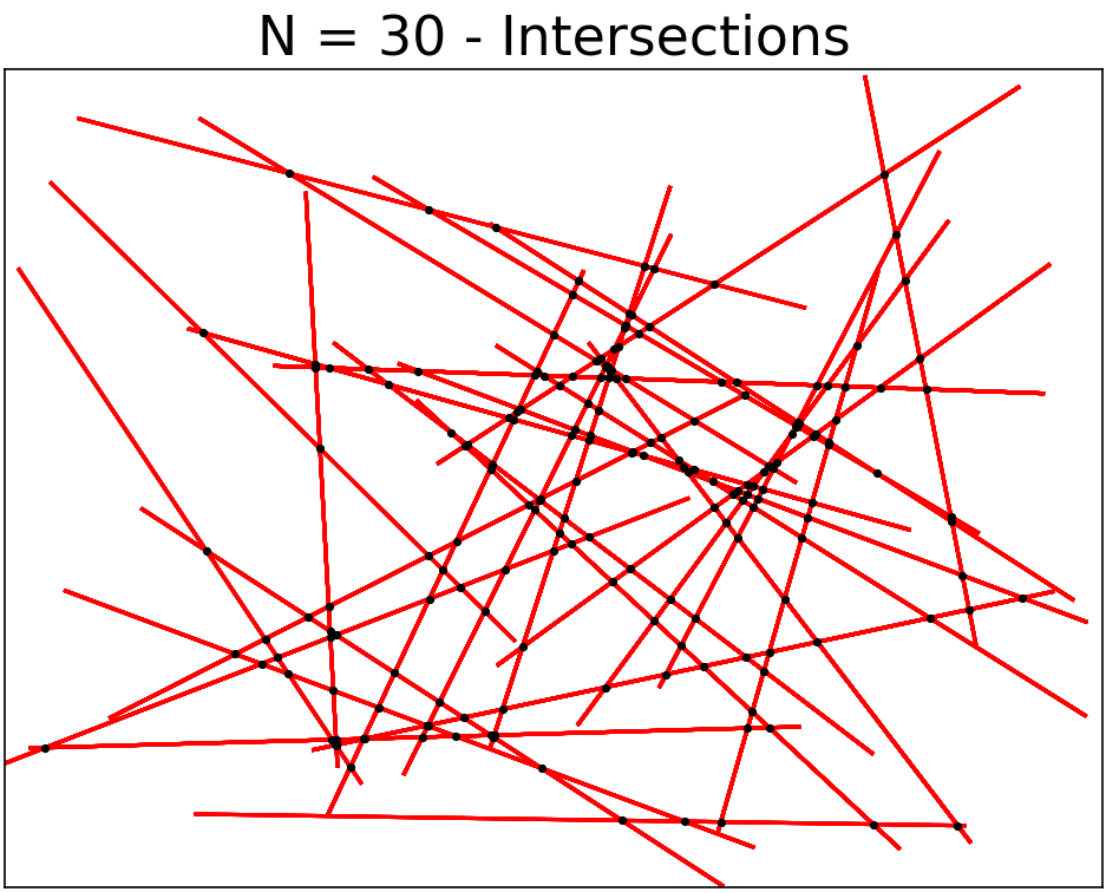


Fig. — 02 intersections N30

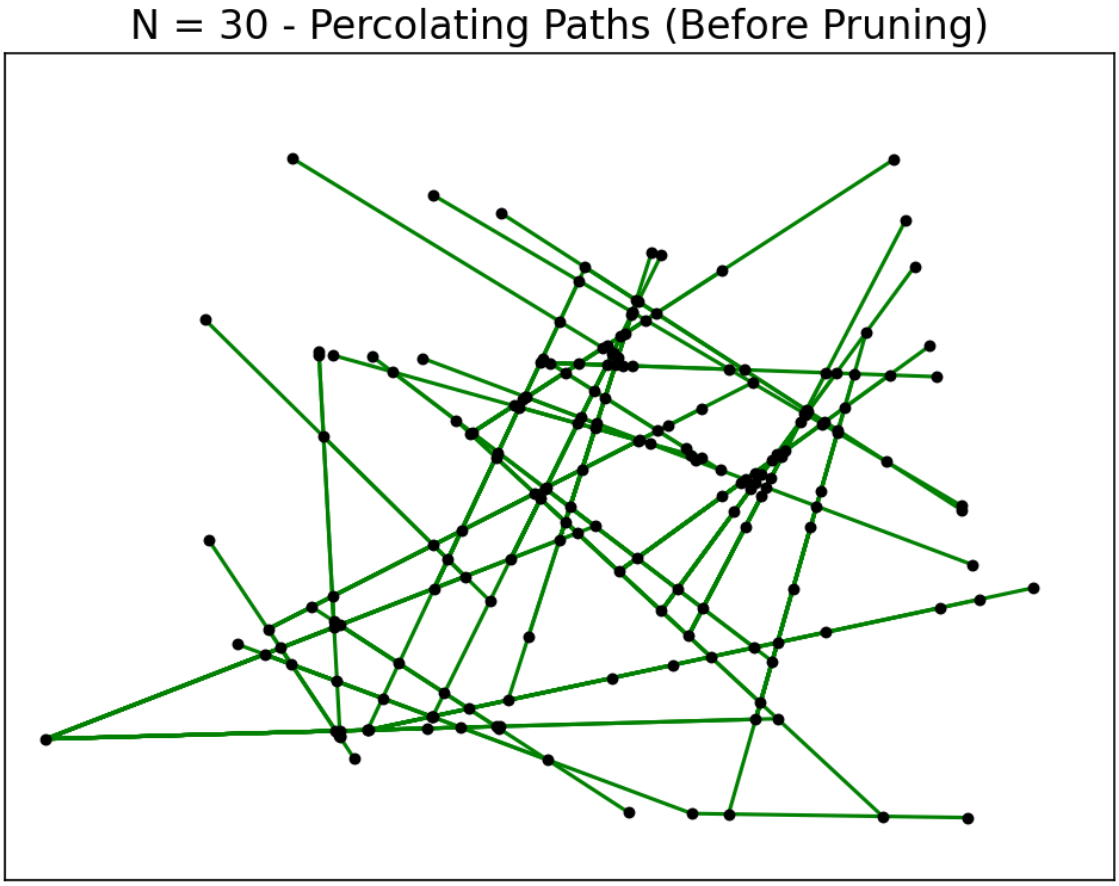


Fig. — 03 percolating paths before pruning N30

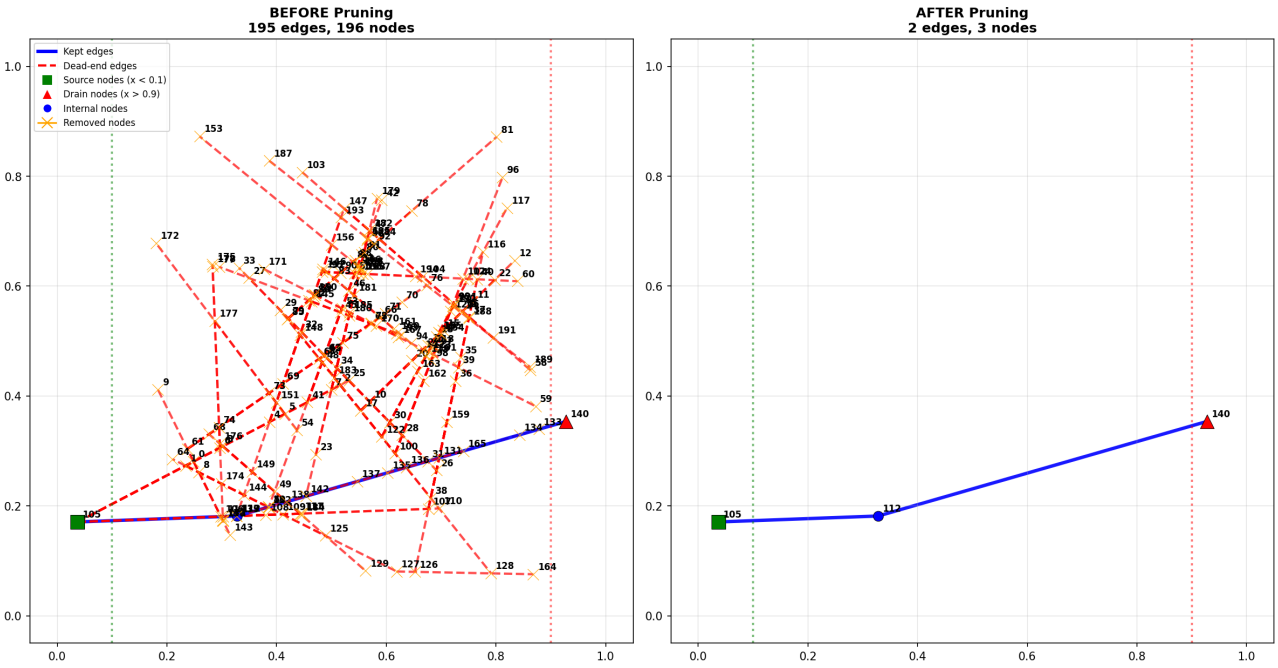


Fig. — 04 before after pruning N30

3. Edge Resistances (Pruned Network)

Edge #	Node 1	Node 2	Conductance (S)	Resistance (Ω)	Voltage Drop (V)
0	105	112	1.0000	1.0000	0.000000
1	112	140	1.0000	1.0000	0.000000

4. Impedance Calculation

Quantity	Value
Applied voltage (V)	1.00 V
Resistance per segment k	1 Ω
Calculated impedance R_eq	2000.000000 Ω
Total current I	5.000000e-04 A

Impedance for different k (resistance per segment) values

k (Ω/segment)	R_equivalent (Ω)
1	2.000000
10	20.000000
100	200.000000
1000	2000.000000
10000	20000.000000